

LeadBoSki: A Smart Personal Assistant for Leadership Support in Video-Meetings

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ABSTRACT

Leading teams effectively is an essential factor for team performance. Main events where leaders practice leadership are video-meetings which are core elements of remote teamwork. However, leading teams is challenging. In particular, in video-meetings, leadership is complex due to limitations such as aggravated understanding of social signals and team dynamics. Voice-based smart personal assistants (SPA) such as Amazon's Alexa can provide real-time support and have the potential to maintain effective leadership in video-meetings. Therefore, we developed the SPA LeadBoSki for real-time leadership support in video-meetings. LeadBoSki detects leadership potentials in video-meetings and sends leadership potential advice to the leader. Pilot study results with two real-world teams ($n=9$) show a positive leader experience and stimulating behavior by the leader for the team members' motivation. Our work contributes with the SPA LeadBoSki as the first SPA for leadership support in video-meetings of its kind and as a potential use-case for future workplaces.

CCS CONCEPTS

• **Human-centered computing** → **Sound-based input / output**; *Interactive systems and tools*; **Natural language interfaces**.

KEYWORDS

Transformational leadership, Smart personal assistant, Voice assistant, Remote team, Video-meeting

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1 INTRODUCTION

Leadership is an important skill and essential factor for team performance [5]. However, not everybody is equally capable of effectively guiding teams since leadership is complex [9]. There are multiple ways leaders apply leadership (e.g., in one-on-one feedback sessions, in speeches, or via non-verbal behavior). With technological progress as well as the COVID-19 pandemic, probably the most common occasion where leaders practice leadership are video-meetings [17]. Such meetings, where all members assemble, have unique character and impact since every member experiences the team leader.

However, in video-meetings the interpretation of social signals of others and team dynamics is challenging [4]. This limitation aggravates effective leadership [9]. While effective leadership has highly positive effects not only on team performance but also on individual well-being, bad leadership may have negative consequences such as team conflict or higher individual stress [14]. To maintain effective leadership, real-time support interventions are one possible solution for the leader to immediately counteract these challenges [14] and, therefore, is desirable.

We propose to leverage voice-based smart personal assistants (SPA), such as Amazon's Alexa, as video-meeting assistants for dynamic leadership support. SPAs interact via voice and natural language with their users, they *speak*. The ability of SPAs to speak with the team leader allows effective leadership interventions. Through their scalability, accessibility, as well as natural language capabilities, SPAs are particularly suited for supporting team leaders via voice during video-meetings [21]. While voice-based interventions also bring uncertainties, such as the risk of interruptions or the transparency of the content, we believe that with user-centered design, the benefits outweigh its risks. Therefore, in this study, we present the SPA LeadBoSki as a leadership support system for video-meetings. LeadBoSki detects leadership potential during video-meetings through a dialogue act detection engine [16] and sends leadership potential advice to the leader.

We evaluated LeadBoSki in a pilot study with two real-world teams ($n_{participants} = 9$). Qualitative results show a positive experience with LeadBoSki, appraisal of the concrete advice by the leaders, and a more detailed leader's interest perceived by the team members. With this study, we contribute with first findings and with LeadBoSki as voice-based SPA for leadership support in video-meetings as the first prototype of its kind.

2 BACKGROUND

Transformational Leadership in Teams. Leadership in teams "[...] is a social and goal-oriented influence process" (p.1727) [7]. In this process, leadership behavior affects outcomes such as team performance through follower motivation [2, 7]. Transformational leadership (TL) posits that exceptional performance is created by a mission and new ways of thinking and learning [8]. Transformational leaders activate followers' social identities and encourage people to collaborate [15]. TL is characterized by four components: idealized influence (II), which is leadership and motivation by a good example; inspirational motivation (IM), the promotion of enthusiasm and energy in followers; intellectual stimulation (IS), the inspiration of followers to develop their own innovative strategies and challenge standards; and individualized consideration (IC), the establishment of a supportive climate and mentoring to raise followers' personal abilities [15].

Smart Personal Assistants for Team Leadership. Voice-based smart personal assistants (SPA), chatbots, or smart speakers are software-based systems designed to interact with humans via voice in- and output using natural language [6, 11, 22]. Recently, research has increasingly applied SPAs in teams and adopted for personal development support, as well as team and task management, often in work or educational context [12, 18]. For example, [22] or [20] applied Amazon's Alexa as a voice-based SPA for collaborative problem-solving and tutoring in teams. [21] successfully built a voice-based SPA in order to tutor people with public speaking anxiety. These examples showcase the general uprising relevance of the topic, as well as the effectiveness and potential of SPAs in teams. However, [19] have reviewed the potential for chatbots in groups and identified that chatbots as team leaders have not been explored yet. Hence, to the best of our knowledge, there are no studies of SPAs for leadership support in teams so far, in particular not in video-meetings which represent an essential element of the remote workplace.

3 LEADBOSKI - A LEADERSHIP SKILL FOR AMAZON'S ALEXA PLATFORM

We created LeadBoSki as voice-based SPA that identifies leadership potentials through natural language processing and dialogue act detection. LeadBoSki is designed as a specific application on the Amazon Alexa platform, an Alexa skill. Based on its abilities it informs the leader in video-meetings about leadership improvement potentials. This support is only experienced by the leader in order to avoid exposing the leader's identified potentials to the team. With regard to privacy protection, LeadBoSki is designed so that recording, transcription, and identification of leadership support only includes the leader's voice and no other team members.

3.1 LeadBoSki Experience Design in Video-Meetings

LeadBoSki is situated on site of the leader during video-meetings in form of an Alexa Echo device. The interaction flow of LeadBoSki is described in Figure 1. Before the meeting, the leader starts LeadBoSki through a voice command. During the meeting, LeadBoSki actively listens to the leader and analyzes the voice in real-time. In

essence, the LeadBoSki engine identifies dialogue acts (DA) within leadership behavior through dynamic DA tagging [16]. The DAs are mapped to TL leadership characteristics and the system analyzes the sentiment of the leader's voice. Through the identification of leadership patterns and negative sentiment, LeadBoSki detects leadership improvement potentials. In case of positive detection, LeadBoSki notices the leader after a no-intervention period ($p_{minimum} = 5$ min) in which no intervention of LeadBoSki takes place. The leader addresses LeadBoSki through a command and LeadBoSki provides a short comprehension about leadership potential and an example on how to address it. Since the SPA is situated at the leader's location, the interaction is experienced by the leader only.

LeadBoSki is able to address three different leadership potentials: (1) the potential for the TL ability IC (individualized consideration) for requesting team member feedback; for (2) the TL ability IM (inspirational motivation) for presenting an optimistic outlook; and (3) for the TL ability IS (intellectual stimulation) for stimulating team members' thoughts.

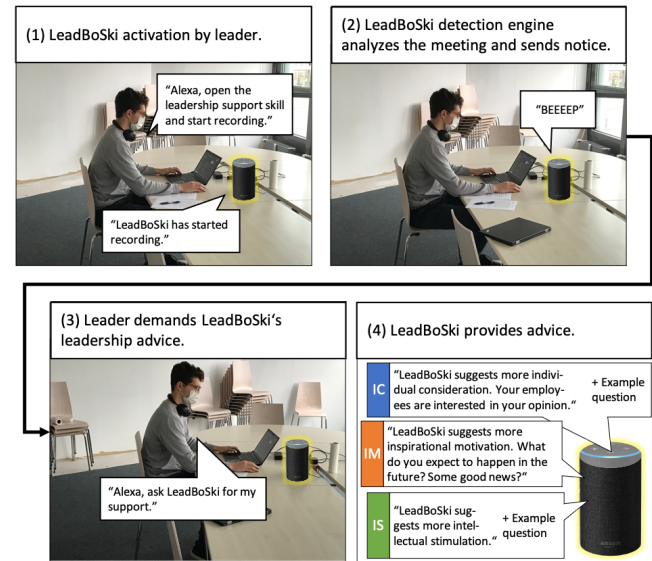


Figure 1: LeadBoSki interaction flow from activation to the reception of leadership advice.

3.2 LeadBoSki System

We implemented LeadBoSki as a program for the Amazon Alexa platform (as Alexa skill). Following [21] we chose Alexa as SPA due to its effective speech modality, high user adherence, and accessibility for the leader. Through its SPA nature, LeadBoSki may be received as a team facilitator. It is developed based on the Alexa Skill Kit SDK for node.js and deployed through Amazon Lambda. Since a continuous analysis of meeting voice is not possible via Alexa, we developed a python-based Flask server [1] locally that contains the recording functionality through the user's microphone, the leadership potential detection engine, and a Rest API to expose detection results to the LeadBoSki Alexa skill.

3.2.1 LeadBoSki Detection Engine & Logic. The LeadBoSki detection engine contains the (1) processing of the leader’s voice, the (2) leader’s utterance analysis, and the (3) leadership potential detection (see Figure 2). First, the engine receives the speech via the user’s microphone, transcribes the speech to text, and identifies individual speaker’s utterances by using the Google Cloud Speech-to-Text API. Thus, the voice is processed in real-time and there is long-time storage of records. Second, in the analysis step, three classifications take place. LeadBoSki classifies the spoken utterances based on the domain- and task-independent DAs [16]. DAs are speech or communicative acts in human conversations organized in a taxonomy. For example, first-level categories of the DA taxonomy are *General* for general statements or questions, or *Social obligation management* for thanking or salutations. Based on this, [16] have developed a large domain- and task-independent body of DA tagging training data on which LeadBoSki implements the proposed machine learning DA tagging model. For this system, it tags a DA for each speech utterance. The DA tagger yields an accuracy of between 71.3% and 86.6% [16]. In parallel, each utterance content is analyzed by the sentiment analysis tool VADER [10] which results in a sentiment value per utterance, and a rule-based classifier for words that reference to future action. This allows analyzing spoken words directed towards future action. In the third step, LeadBoSki identifies leadership potentials based on a TL potential detection logic. In the positive case of detection, LeadBoSki sends out one of the three leadership potentials in form of a potential notice.

The TL detection logic is able to detect three of the four TL features. TL potential for the feature *IC* (individualized consideration) is detected if the DA category *Feedback* is identified with a sentiment score $s_i > 0$ (*IC* describes the inclusion of team members’ perspectives into leader reflections [13]). The feature *IM* (inspirational motivation) applies if the DA category *Statement* with a sentiment score $s_i > 0$ and utterance with future focus is detected in advance (*IM* describes the provision of a vision statement with an optimistic outlook). The feature *IS* (intellectual stimulation) applies if a DA *Directive* or *SetQ* is detected in the previous utterance (maps to the member stimulation to reconsider challenges). To counteract repeated interruption of the meeting a period of no intervention ($p_{minimum} = 5 \text{ min}$) is applied. We assessed $p_{minimum}$ based on multiple pre-tests.

4 PILOT STUDY

To investigate how the leader and the team experience LeadBoSki we conducted a pilot study. We recruited two real-world teams ($n_{participants} = 9$), that regularly hold video-meetings and have one leader and multiple team members. One team worked on a technical development, the other one on a business consultancy project, and the video meetings were the weekly update meetings. On site of the leader, an Amazon Alexa Echo device was situated. After a short introduction by the experimenters, the leaders started LeadBoSki and the video-meeting began. The team members were informed about the presence and the general function (e.g., voice processing) of LeadBoSki and signed an agreement form, while they did not know the intervention form and content. During the meeting, the leaders had no restrictions and all participants behaved as usual. The meetings lasted between 45-80 minutes. In

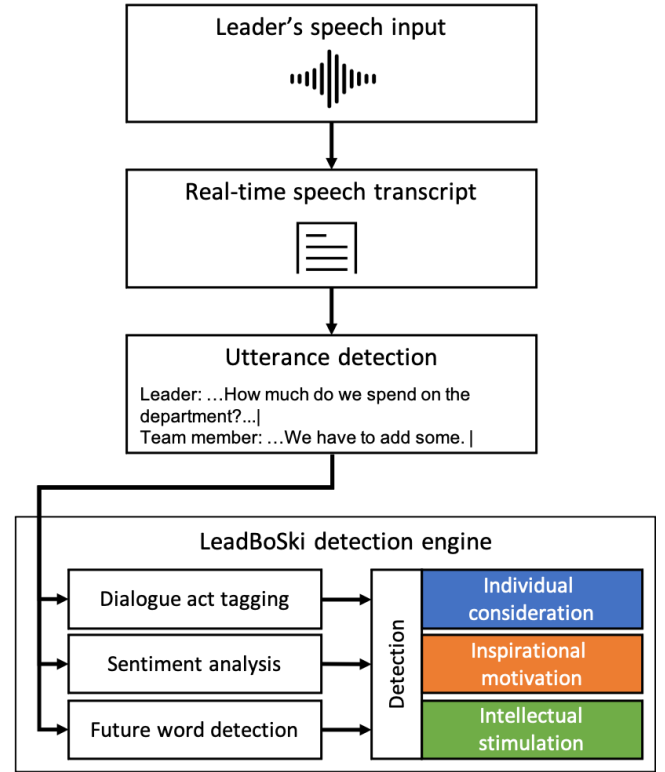


Figure 2: LeadBoSki detection engine with detection logic.

both meetings, LeadBoSki intervened multiple times and the leaders asked for advice. Also, all leadership potential categories were raised at least once by LeadBoSki. After the meeting, we conducted semi-structured, qualitative interviews with all participants. Two different interview guidelines were developed, one for the leader with detailed questions about the leadership support, and one for the team members which focused on the perception of the leader by the team members. An example question for the leader guideline was "How did you perceive the effect of your leadership feedback on the team?". The interviews were then transcribed and analyzed deductively following the methodology of [3].

4.1 Pilot Study Results

4.1.1 Leader Perspective. LeadBoSki received positive feedback from the leaders. Both leaders (L1, L2) mentioned that the interventions by LeadBoSki helped them with a perceived positive influence on leadership and team interaction. L1 documented "I actually thought it was pretty cool that you had support like this.". The first notification of the acoustic signal already allowed the leaders to reconsider their behavior.

Feature effect. The *IC* leadership potential advice was perceived as positive: e.g., L2 stated "[...] what I really liked was when LeadBoSki proposed to go more into detail about the opinions of the team members.". Also, there was an indirect positive influence on the team since the leaders perceived that the members liked the stimulation by the leader’s behavior. Existing research states that TL in remote scenarios is challenging [15]. Therefore, the results of leadership

enhancement through LeadBoSki in video-meetings are especially valuable.

Intervention trigger. As a possible reason for the leadership improvement, L2 cited the link to the intervention trigger of the IC leadership potential advice, which increased the credibility of LeadBoSki. Thereby, the leaders included members' thoughts stronger in the meetings which stimulated team spirit. In turn, for the two features IM and IS the leaders could not derive the cause of intervention. L2 reported a discussion about technical details, where "[...] a vision statement would not be appropriate, since these topics refer to two different levels. Here, I would be interested why an intervention took place.". A reason for this might be the meeting context. For example, meeting two had a software development focus, where a visionary statement might be problematic. Consequently, LeadBoSki's detection logic needs to include this factor. As positive aspect, leaders mentioned the actionable example provided which guided their ad-hoc reaction.

Behavioral adaption. Both leaders documented a behavioral adaptation during the meeting after interacting with LeadBoSki. While both listened to the whole advice in the beginning, after a while they focused only on the acoustic signal and the leadership potential type.

Improvement potential. Everybody documented the need to divide the attention between the video-meeting and LeadBoSki's advice. This led to a short meeting interruption. L1 reported, "*The team gets confused shortly and thinks to itself, now he's talking about something that we don't know [...]. So parallel is difficult.*". Both proposed to show the advice in text form.

4.1.2 Team Member Perspective. Team members (M1.1-3, M2.1-4) reported a positive experience with slight differences to regular meetings since short periods were mentioned where the leader focused on LeadBoSki. This indicates that LeadBoSki did not lead to negative interference which is a positive signal for voice-based SPAs in video-meetings. For several participants, the leader was assessed as positive as usual or better. They reported that the leader went more into detail and engaged more with the members' opinions. M1.2 reported, "[...] the leader incorporated feedback that came from us as a project team and implemented that in this meeting."

5 CONCLUSION AND FUTURE WORK

In this paper, we introduced LeadBoSki, a voice-based SPA for leadership support in video-meetings. LeadBoSki gives dynamic improvement potential advice for team leaders. Results from our pilot study with two teams showed appreciation of LeadBoSki's leadership support and stimulation of team members' motivation.

In future work, we plan to address the limitations of multitasking for leaders and expand LeadBoSki with an additional modality of textual presentation. Second, the value of LeadBoSki will expand with continuous usage. While, there are potential workplace barriers to overcome (e.g., workplace regulations for audio recording), we believe in its value and plan to conduct a longitudinal field study with multiple teams. Our work supports previous findings from TL research and showcases the applicability of voice-based SPAs for video-meeting support in general and leadership support in particular. We contribute with LeadBoSki as the first

SPA for video-meeting support as a potential use-case for future workplaces.

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